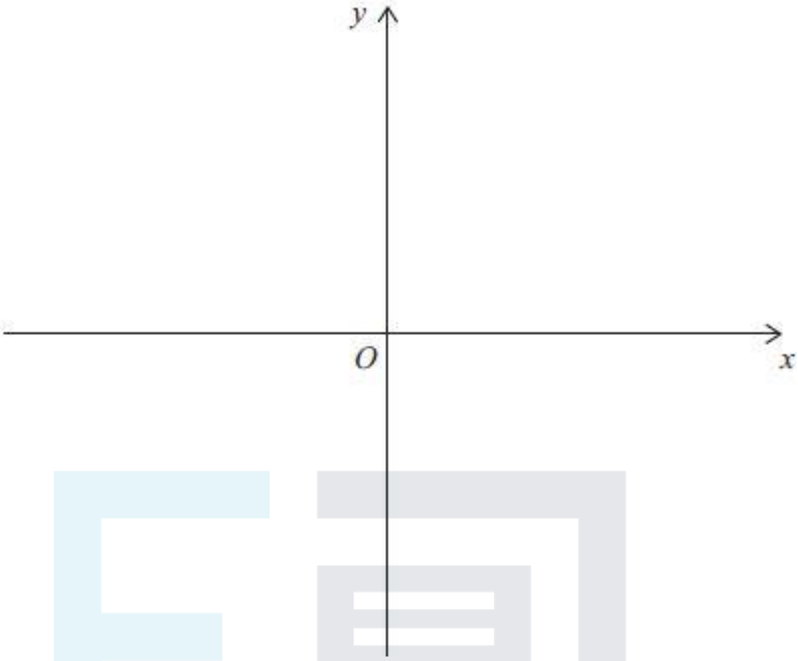
Q1.

(a) Sketch the graph of $y^2 = 4x$



(1)

(b) Ben is using a 3D printer to make a plastic bowl which holds exactly 1000 cm^3 of water.

Ben models the bowl as a region which is rotated through 2π radians about the x-axis.

He uses the finite region enclosed by the lines $x = d$ and $y = 0$ and the curve with equation $y^2 = 4x$ for $y \geq 0$

(i) Find the depth of the bowl to the nearest millimetre.

(4)

(ii) What assumption has Ben made about the bowl?

(1)

(Total 6 marks)

Q2.

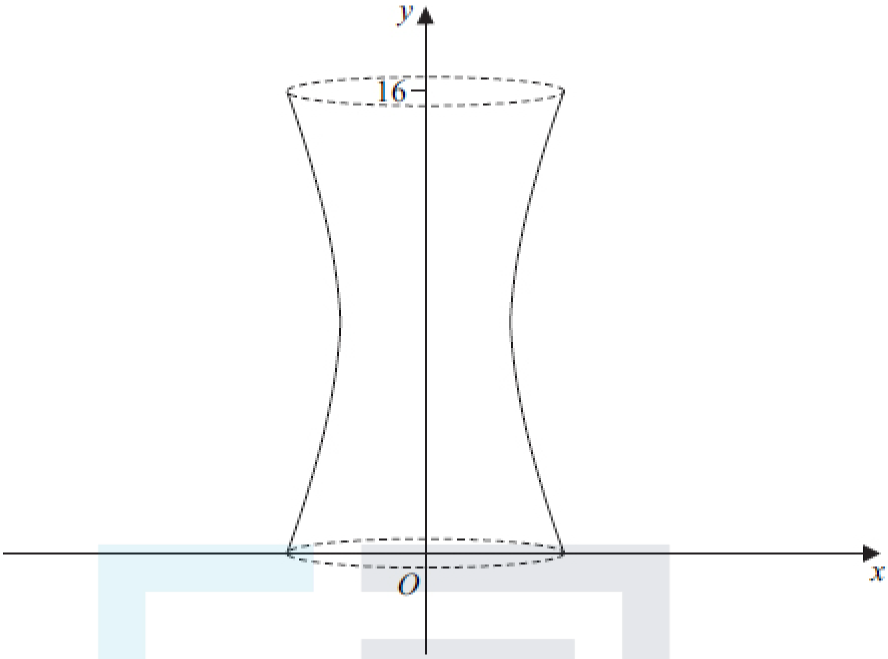
The region bounded by the curve with equation $y = 3 + \sqrt{x}$, the x-axis and the lines $x = 1$ and $x = 4$ is rotated through 2π radians about the x-axis.

Use integration to show that the volume generated is $\dfrac{125\pi}{2}$

(Total 5 marks)

Q3.

The shape of a vase can be modelled by rotating the curve with equation $16x^2 - (y - 8)^2 = 32$ between $y = 0$ and $y = 16$ completely **about the y-axis**.



The vase has a base.

Find the volume of water needed to fill the vase, giving your answer as an exact value.

(Total 5 marks)

Q4.

(a) A curve has equation

$y = x^2 e^{-\frac{x}{4}}$.

Show that the curve has exactly two stationary points and find the exact values of their coordinates.

(7)

(b) (i) Use integration by parts twice to find the exact value of $\int_0^4 x^2 e^{-\frac{x}{4}} \, dx$.

(7)

(ii) The region bounded by the curve $y = 3xe^{-\frac{x}{8}}$, the x-axis from 0 to 4 and the line $x = 4$ is rotated through 360° about the x-axis to form a solid.

Use your answer to part (b)(i) to find the exact value of the volume of the solid generated.

(2)

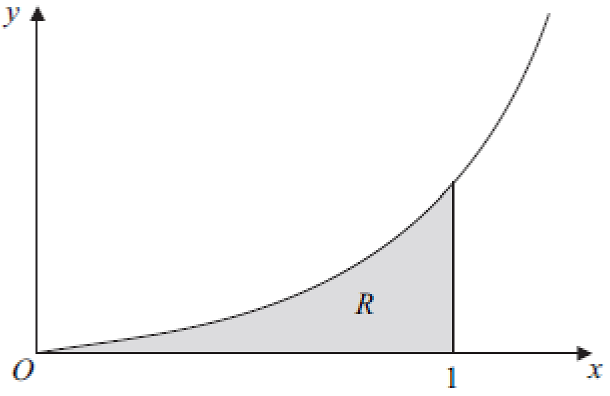
(Total 16 marks)

Q5.

(a) By using integration by parts, find $\int x \, e^{6x} \, dx$.

(4)

(b) The diagram shows part of the curve with equation $y = \sqrt{x}\, e^{3x}$.



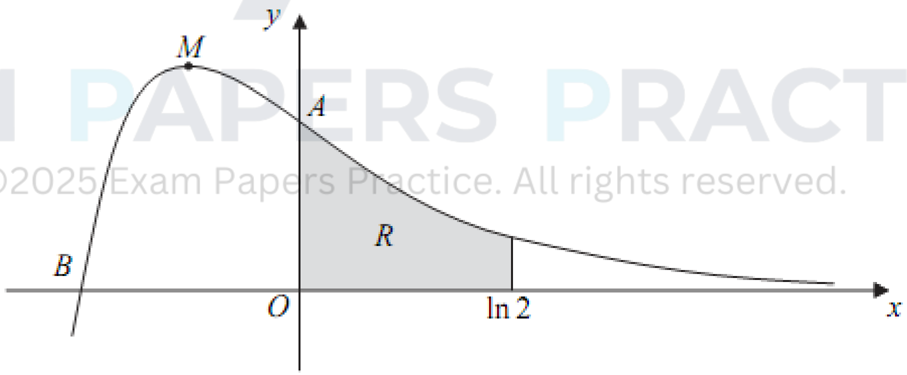
The shaded region R is bounded by the curve $y = \sqrt{x}\, e^{3x}$, the line $x = 1$ and the x-axis from $x = 0$ to $x = 1$.

Find the volume of the solid generated when the region R is rotated through 360° about the x-axis, giving your answer in the form $\pi(pe^6 + q)$, where p and q are rational numbers.

(3)
(Total 7 marks)

Q6.

The diagram shows the curve $y = 4e^{-2x} - e^{-4x}$.



The curve crosses the y-axis at the point A, the x-axis at the point B, and has a stationary point at M.

The shaded region R is bounded by the curve $y = 4e^{-2x} - e^{-4x}$, the lines $x = 0$ and $x = \ln 2$ and the x-axis.

Find the volume of the solid generated when the region R is rotated through 360° about the x-axis, giving your answer in the form $\frac{p}{q}\pi$, where p and q are integers.

(Total 7 marks)

Q7.

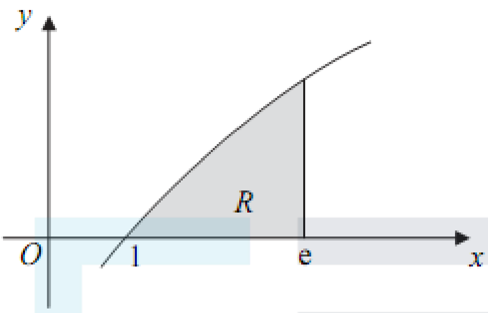
(a) Use integration by parts to find $\int x \ln x \, dx$.

(3)

(b) Given that $y = (\ln x)^2$, find $\frac{dy}{dx}$.

(2)

(c) The diagram shows part of the curve with equation $y = \sqrt{x} \ln x$.



The shaded region R is bounded by the curve $y = \sqrt{x} \ln x$, the line $x = e$ and the x-axis from $x = 1$ to $x = e$.

Find the volume of the solid generated when the region R is rotated through 360° about the x-axis, giving your answer in an exact form.

(6)

(Total 11 marks)

Q8.

(a) Use the mid-ordinate rule with four strips to find an estimate for $\int_0^{12} \ln(x^2 + 5)\,dx$, giving your answer to three significant figures.

(4)

(b) A curve has equation $y = \ln(x^2 + 5)$.

(i) Show that this equation can be rewritten as $x^2 = e^y - 5$.

(1)

(ii) The region bounded by the curve, the lines $y = 5$ and $y = 10$ and the y-axis is rotated through 360° about the y-axis. Find the exact value of the volume of the solid generated.

(4)

(Total 9 marks)

Q9.

(a) Use integration by parts to find:

(i) $\int x\cos 4x \, dx$;

(4)

(ii) $\int x^2 \sin 4x \, dx$.

(4)

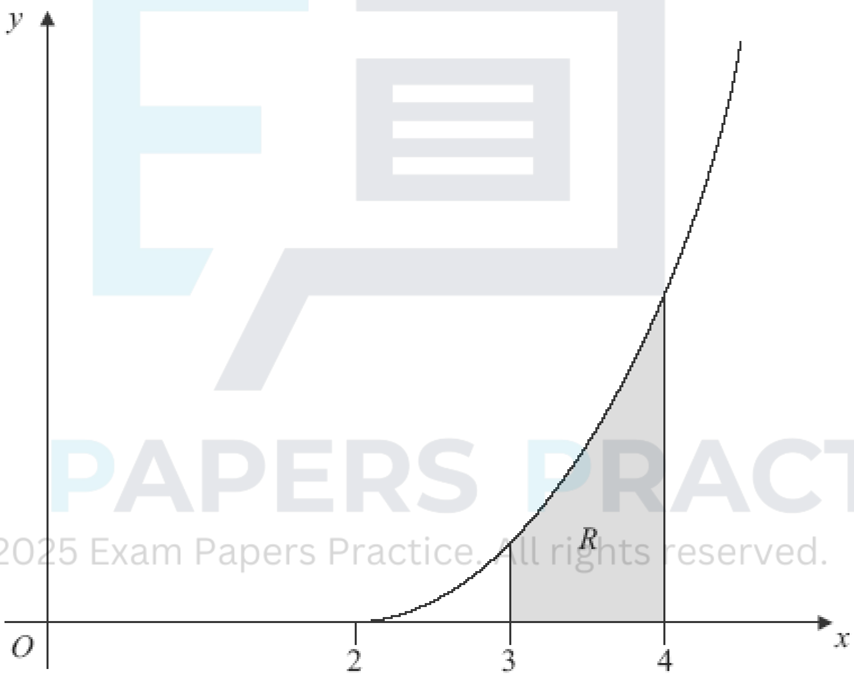
(b) The region bounded by the curve $y = 8x\sqrt{(\sin 4x)}$ and the lines $x = 0$ and $x = 0.2$ is rotated through 2π radians about the x-axis. Find the value of the volume of the solid generated, giving your answer to three significant figures.

(3)

(Total 11 marks)

Q10.

The diagram shows the curve with equation $y = \sqrt{(x-2)^5}$ for $x \geq 2$.



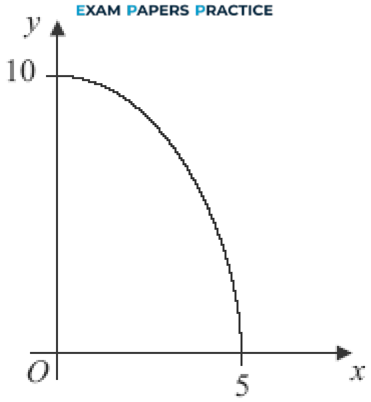
The shaded region R is bounded by the curve $y = \sqrt{(x-2)^5}$, the x-axis and the lines $x = 3$ and $x = 4$.

Find the exact value of the volume of the solid formed when the region R is rotated through 360° about the x-axis.

(Total 4 marks)

Q11.

The diagram shows the curve with equation $y = \sqrt{100 - 4x^2}$, where $x \geq 0$.



(a) Calculate the volume of the solid generated when the region bounded by the curve shown above and the coordinate axes is rotated through 360° about the ***y*-axis**, giving your answer in terms of π.

(5)

(b) Use the mid-ordinate rule with five strips of equal width to find an estimate for $\int_0^5 \sqrt{100-4x^2}\,dx$, giving your answer to three significant figures.

(4)

(Total 9 marks)

Q12.

(a) Given that $e^{-2x} = 3$, find the exact value of x.

(2)

(b) Use integration by parts to find $\int xe^{-2x}\,dx$.

(4)

(c) A curve has equation $y = e^{-2x} + 6x$.

(i) Find the exact values of the coordinates of the stationary point of the curve.

(4)

(ii) Determine the nature of the stationary point.

(2)

(iii) The region R is bounded by the curve $y = e^{-2x} + 6x$, the x-axis and the lines $x = 0$ and $x = 1$.

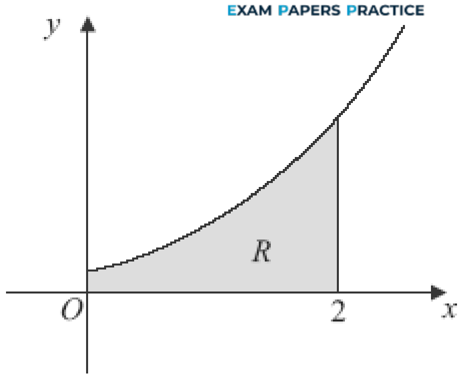
Find the volume of the solid formed when R is rotated through 2π radians about the x-axis, giving your answer to three significant figures.

(5)

(Total 17 marks)

Q13.

The diagram shows the curve with equation $y = (e^{3x} + 1)^{\frac{1}{2}}$ for $x \geq 0$.



(a) Find the gradient of the curve $y = (e^{3x} + 1)^{\frac{1}{2}}$ at the point where $x = \ln 2$.

(5)

(b) Use the mid-ordinate rule with four strips to find an estimate for $\int_0^2 (e^{3x} + 1)^{\frac{1}{2}} dx$, giving your answer to three significant figures.

(4)

(c) The shaded region R is bounded by the curve, the lines $x = 0$, $x = 2$ and the x-axis. Find the exact value of the volume of the solid generated when the region R is rotated through 360° about the x-axis.

(4)

(Total 13 marks)

Q14.

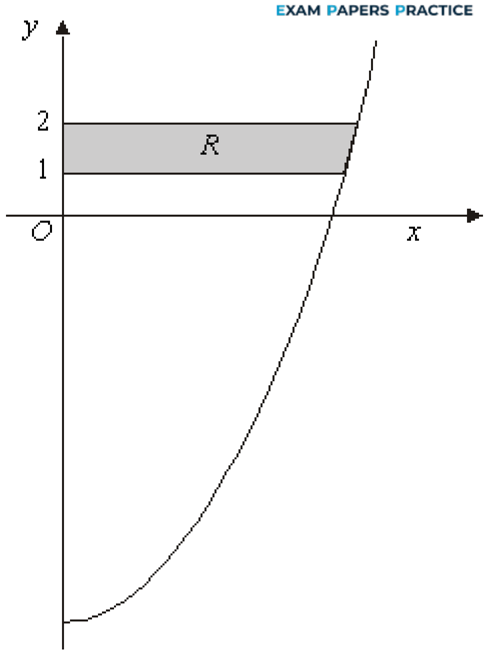
(a) Use integration by parts to find $\int x \sin x \, dx$.

(4)

(b) Using the substitution $u = x^2 + 5$, or otherwise, find $\int x\sqrt{x^2 + 5} \, dx$.

(4)

(c) The diagram shows the curve $y = x^2 - 9$ for $x \geq 0$.



The shaded region R is bounded by the curve, the lines $y = 1$ and $y = 2$, and the y-axis.

Find the exact value of the volume of the solid generated when the region R is rotated through 360° **about the y-axis**.

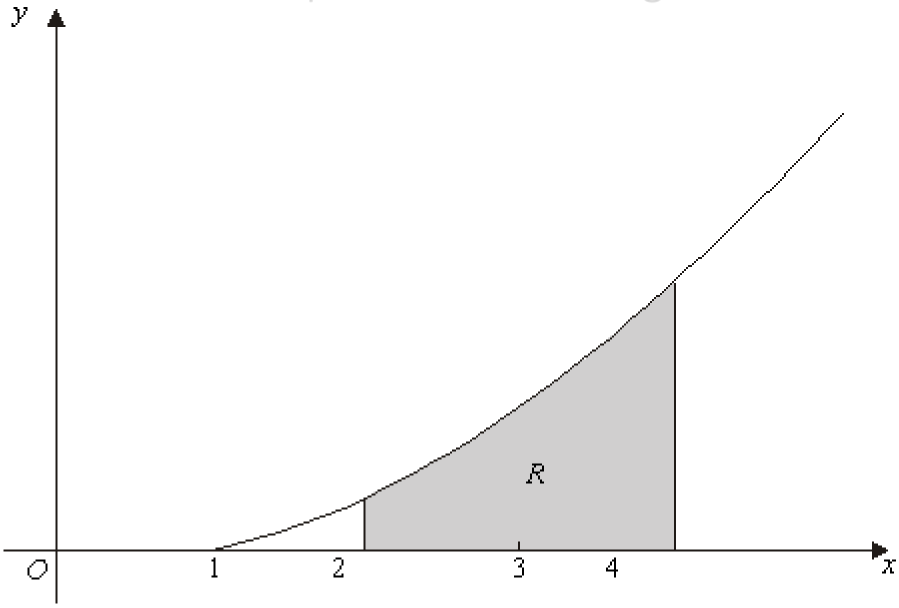
(4)
(Total 12 marks)

Q15.

(a) Differentiate $(x - 1)^4$ with respect to x.

(1)

(b) The diagram shows the curve with equation $y = 2\sqrt{(x-1)^3}$ for $x \geq 1$.



The shaded region R is bounded by the curve $y = 2\sqrt{(x-1)^3}$, the lines $x = 2$ and $x = 4$, and the x-axis.

Find the exact value of the volume of the solid formed when the region R is rotated through 360° about the x-axis.

(4)

(c) Describe a sequence of **two** geometrical transformations that maps the graph of $y = \sqrt{x^3}$ onto the graph of $y = 2\sqrt{(x-1)^3}$

(4)
(Total 9 marks)

Q16.

(a) Given that $z = \frac{\sin x}{\cos x}$, use the quotient rule to show that $\frac{dz}{dx} = \sec^2 x$.

(3)

(b) Sketch the curve with equation $y = \sec x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

(2)

(c) The region R is bounded by the curve $y = \sec x$, the x-axis and the lines $x = 0$ and $x = 1$.

Find the volume of the solid formed when R is rotated through 2π radians about the x-axis, giving your answer to three significant figures.

(3)
(Total 8 marks)

Q17.

(a) (i) Sketch the graph of $y = 4 - x^2$, indicating the coordinates of the points where the graph crosses the coordinate axes.

(2)

(ii) The region between the graph and the x-axis from $x = 0$ to $x = 2$ is rotated through 360° about the x-axis. Find the exact value of the volume of the solid generated.

(4)

(b) (i) Sketch the graph of $y = |4 - x^2|$.

(2)

(ii) Solve $|4 - x^2| = 3$.

(3)

(iii) Hence, or otherwise, solve the inequality $|4 - x^2| < 3$.

(2)
(Total 13 marks)